

Zihan Yang

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About Me

I am a master's student at Beihang University with a background in materials science and aerospace engineering, now focusing on robotics. My research interests lie at the intersection of control and machine learning, covering both learning-based planning and control, as well as control-theoretic methods that improve learning algorithms. Recent works have been recognized at ICLR 2025 (Oral) and ICCA 2024 (Best Student Paper Award).

Education

Northwestern Polytechnical University, B.Eng. in Material Science and Engineering	Sept 2019 - June 2023
Queen Mary University of London B.Eng. in Material Science and Engineering (UK-China Transnational Education)	Sept 2019 - June 2023
• GPA: 3.6/4.0	
• First Class Honors.	
Beihang University, M.S. in Aerospace Science and Technology	Sept 2023 - June 2026 (expected)
• GPA: 3.8/4.0, supervised by Prof. Kexin Guo, Prof. Xiang Yu, and Prof. Lei Guo	

Publications

(* equal contribution)

- [1] **Z. Yang**, J. Jia, M. Wang, Y. Liu, K. Guo, and X. Yu, "Unified meta-representation and feedback calibration for general disturbance estimation," *Under Review*, 2025.
- [2] J. Jia*, M. Wang*, **Z. Yang**, B. Yang, Y. Liu, K. Guo, and X. Yu, "Learning-based observer for coupled disturbance," 2025.
- [3] J. Jia*, **Z. Yang***, M. Wang, K. Guo, J. Yang, X. Yu, and L. Guo, "Feedback favors the generalization of neural ODEs," in *The Thirteenth International Conference on Learning Representations*, **Oral Presentation (acceptance rate: 1.8%)**, 2025.
- [4] K. Guo*, **Z. Yang***, J. Jia, Y. Liu, and X. Yu, "Optimizing control-friendly trajectories with self-supervised residual learning," *IEEE Transactions on Automation Science and Engineering*, 2025.
- [5] **Z. Yang**, J. Jia, Y. Liu, K. Guo, X. Yu, and L. Guo, "TRACE: Trajectory refinement with control error enables safe and accurate maneuvers," in *IEEE 18th International Conference on Control and Automation*, **Best Student Paper Award**, 2024.
- [6] X. Song, X. Guo, G. Liu, **Z. Yang**, and L. Liu, "Co-optimization of motion and energy domain for hydrogen-powered hybrid UAVs: A bi-directional coupling architecture," in *IEEE 19th International Conference on Control and Automation*, 2025.
- [7] K. Guo, Y. Liu, J. Jia, **Z. Yang**, S. Zhou, X. Yu, and L. Guo, "Flying in harsh environments: Anti-rain quantification system and control strategy," *IEEE Transactions on Industrial Electronics*, 2025.

Talks

Presentation IEEE International Conference on Control and Automation (ICCA), Reykjavík, Iceland.	June, 2024
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Internship

COMAC (Commercial Aircraft Corporation of China, Ltd.) Beijing Aircraft Technology Research Institute, Intern	July, 2022 - August, 2022
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Honors and Awards

Oral Presentation, Top 1.8% The Thirteenth International Conference on Learning Representations (ICLR), Singapore. April, 2025

Best Student Paper Award *IEEE* International Conference on Control and Automation (ICCA), Reykjavík, Iceland. June, 2024

Funds

Second-Class Scholarship Beihang University, Beijing, China. 2024

Second-Class Scholarship Beihang University, Beijing, China. 2023

Skills

Coding: Python (skilled), Matlab/Simulink (skilled), C/C++ (familiar), HTML/CSS (basic)

Hardware: Skilled in UAV hardware-software co-design and system integration; developed fixed-wing and quadrotor platforms with embedded systems (Pixhawk, Jetson Nano, Intel NUC); experienced in rapid prototyping (3D printing, laser cutting), electronic assembly (soldering)...

Language: Mandarin Chinese (native), English (fluent), Japanese (basic)

Misc.: Electric guitar, skiing, badminton, swimming...

Soft Skills.: Self-motivation, critical-thinking, multitasking...